MINOR PROJECT REPORT

# CYBERSECURITY & NETWORKING FUNDAMENTALS

Secure Small Office Network Design, IP Addressing, Subnetting and Troubleshooting

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| **Submitted to** | Unlox Academy |
| **Program** | Cybersecurity Program — Foundation Batch |
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Project objective: demonstrate core cybersecurity concepts, explain how networks communicate, calculate IPv4 subnets, design a segmented office network, and apply a safe, methodical troubleshooting process.

# Contents


• 1. Abstract and Project Objectives

• 2. Cybersecurity Fundamentals

• 3. Networking Fundamentals and Devices

• 4. How the Internet Works

• 5. IP Addressing and Classification

• 6. Subnetting Calculations

• 7. Office Network Design Scenario

• 8. IP Troubleshooting Scenario

• 9. Security Recommendations

• 10. Conclusion and References


Scope note: This report follows the topics and expected submission items in the provided Minor Project Assessment: complete answers, worked subnet calculations, diagrams, scenario reasoning, and correct terminology.

# 1. Abstract and Project Objectives

This project combines cybersecurity and networking fundamentals into one practical case study. It explains the principles used to protect information, describes the devices and protocols that connect computers, calculates IPv4 network ranges, and proposes a basic office network for 45 employees, 5 servers and 10 network devices. A troubleshooting checklist is also included for a computer that can communicate locally but cannot reach an external website.

## Objectives

• Explain the CIA Triad and distinguish vulnerabilities, threats, risks and attacks.

• Describe network types, common network devices, and the path of a web request.

• Identify IPv4 classes and private/public address ranges; compare IPv4 with IPv6.

• Show complete calculations for /24, /26 and a variable-length subnetting design.

• Design a small office network with security controls and a logical layout.

• Troubleshoot loss of internet access in a repeatable, low-risk order.

## Method

The report uses a hypothetical small organization and RFC-style private IPv4 addressing. The proposed design is a learning model, not a production deployment plan; actual implementation should be reviewed by a qualified network administrator.

# 2. Cybersecurity Fundamentals

## Q1. What is cybersecurity?

Cybersecurity is the practice of protecting computers, networks, applications and data from unauthorized access, disruption, alteration or destruction. It matters because organizations depend on accurate information, reliable services and trusted communications. A security incident can cause financial loss, downtime, privacy harm and reputational damage.

## The CIA Triad

| Principle | Meaning | Practical example |
|---|---|---|
| Confidentiality | Only authorized people can view information. | A payroll file is encrypted and accessible only to HR staff. |
| Integrity | Information remains accurate and is changed only in authorized ways. | A system records changes to a customer record and prevents unauthorized edits. |
| Availability | Systems and data are accessible when needed by authorized users. | A business uses backups, updates and redundant internet equipment to reduce downtime. |

## Q2. Vulnerability, threat, risk and attack

| Term | Explanation | Simple example |
|---|---|---|
| Vulnerability | A weakness that could be exploited. | A router still uses its default administrator password. |
| Threat | A person, event or condition that could cause harm. | An attacker scans for routers with default passwords. |
| Risk | The likelihood and impact of a threat exploiting a vulnerability. | The chance and business impact of someone taking over the router. |
| Attack | An action that attempts to exploit a weakness. | Someone attempts to log in using the default password. |

## Q3. Five areas of cybersecurity

• **Network security:** protects network traffic and infrastructure using firewalls, segmentation and secure configurations.

• **Application security:** reduces software weaknesses through secure coding, testing, patching and access controls.

• **Cloud security:** protects cloud-hosted data, identities and services through permissions, encryption and configuration management.

• **Endpoint security:** protects laptops, desktops and servers using updates, anti-malware, disk encryption and device controls.

• **Information security:** protects data throughout its lifecycle with classification, access control, backups and secure disposal.

# 3. Networking Fundamentals and Devices

## Q4. What is networking?

A computer network is a group of interconnected devices that exchange data and share resources using agreed protocols. Organizations use networks to share files and printers, access business applications, communicate, centralize backups, manage devices and reach internet services.

| Type | Coverage | Example |
|---|---|---|
| PAN — Personal Area Network | A few metres around a person | Phone connected to wireless earbuds by Bluetooth. |
| LAN — Local Area Network | A room, home, office or building | Computers and printers connected in one office. |
| MAN — Metropolitan Area Network | A city or metropolitan area | A service provider links branches across a city. |
| WAN — Wide Area Network | Large geographic areas | A company securely connects offices in different states. |

## Q5. Common network devices

| Device | Purpose |
|---|---|
| Hub | Repeats incoming bits to all ports; inefficient and largely obsolete. |
| Switch | Connects devices inside a LAN and forwards Ethernet frames based on MAC addresses. |
| Router | Forwards packets between different IP networks, such as a LAN and the internet. |
| Firewall | Allows or blocks traffic according to security rules. |
| Wireless access point | Allows Wi-Fi devices to join a network and bridges wireless clients to the LAN. |

**Switch vs router:** a switch primarily forwards traffic within a local network using MAC addresses; a router forwards traffic between IP networks using IP addresses and routing information.

## Q6. What happens when a user enters example.com?

• The browser checks whether it can reuse an existing connection or cached DNS result.

• DNS resolves the domain name example.com to an IP address.

• The computer checks whether the destination is on its local subnet. For an external destination, it sends the packet to its default gateway.

• The local network sends frames through the switch and router. The router forwards packets toward the internet; firewall rules may inspect or block the traffic.

• TCP typically establishes a connection for HTTPS, then TLS negotiates encryption and authenticates the server certificate. HTTP sends the request.

• The web server processes the request and returns a response. The browser renders the page. Modern HTTP/3 can use QUIC over UDP instead of TCP.

## Simplified flow diagram

| | | | | |
|---|---|---|---|---|
| **Browser / PC** | → | **DNS resolver** | → | **Web server IP** |
| LAN switch | → | Default gateway / router | → | Internet |



Note: DNS usually resolves names to addresses; it does not itself carry the webpage. The router/gateway moves packets between networks, while the web server supplies the requested content.

# 4. IP Addressing and Classification

## Q7. IPv4 compared with IPv6

| Feature | IPv4 | IPv6 |
|---|---|---|
| Address length | 32 bits | 128 bits |
| Notation | Four decimal octets, e.g. 192.168.1.10 | Hexadecimal groups, e.g. 2001:db8::10 |
| Address capacity | About 4.3 billion addresses before reserved-use considerations | A vastly larger address space |
| Why IPv6? | Limited public address space encouraged techniques such as NAT. | Provides a very large address space and supports modern IP networking. |

## Q8. Class and private/public identification

| Address | Classful range/class | Private or public? | Reason |
|---|---|---|---|
| 10.10.10.5 | Class A (historical classful system) | Private | Within 10.0.0.0/8. |
| 172.16.5.20 | Class B (historical classful system) | Private | Within 172.16.0.0/12 (172.16.0.0–172.31.255.255). |
| 192.168.1.50 | Class C (historical classful system) | Private | Within 192.168.0.0/16. |
| 8.8.8.8 | Class A (historical classful system) | Public | Not in an RFC 1918 private range; commonly used as a public DNS address. |

Important: class A/B/C is a historical classification system. Modern networks use CIDR prefixes such as /24, not class boundaries, to determine network size.

## Q9. Address components for 192.168.20.45/24

| Item | Answer | Working / explanation |
|---|---|---|
| IP address | 192.168.20.45 | Prefix length is /24. |
| Subnet mask | 255.255.255.0 | 24 network bits: 8 + 8 + 8 bits. |
| Network address | 192.168.20.0 | Host bits are set to zero. |
| Host portion | 0.0.0.45 (last octet = 45) | The first 24 bits identify the network; final 8 bits identify the host. |
| Broadcast address | 192.168.20.255 | All 8 host bits are set to 1. |
| Usable host range | 192.168.20.1 – 192.168.20.254 | Network and broadcast addresses are not assigned to ordinary hosts. |
| Usable hosts | 254 | 2^(32–24) – 2 = 256 – 2 = 254. |

# 5. Subnetting Calculations

## Q10. Basic subnet calculation: 192.168.10.0/26

A /26 prefix leaves 32 – 26 = 6 host bits. The subnet mask is 255.255.255.192, because the final mask octet is 11000000 in binary (192). Each subnet contains 2■ = 64 total addresses and 64 – 2 = 62 usable host addresses.

| Requested item | Answer | Working |
|---|---|---|
| Subnet mask | 255.255.255.192 | /26 = 26 network bits. |
| Number of /26 subnets from /24 | 4 | Borrowed bits = 26 – 24 = 2; $2^2 = 4$. |
| Usable hosts per subnet | 62 | 2■ – 2 = 62. |
| Network address | 192.168.10.0 | First block starts at .0. |
| First usable | 192.168.10.1 | Network + 1. |
| Last usable | 192.168.10.62 | Broadcast – 1. |
| Broadcast address | 192.168.10.63 | Block size 256 – 192 = 64; first block spans .0–.63. |

The four /26 networks inside 192.168.10.0/24 are 192.168.10.0/26, 192.168.10.64/26, 192.168.10.128/26 and 192.168.10.192/26.

## Q11. Variable-length subnetting design

Requirement: IT 50 hosts, HR 30, Finance 20 and Management 10. Allocate the largest requirement first. A /26 supports 62 usable hosts, /27 supports 30, /27 supports 30, and /28 supports 14. These four networks use 64 + 32 + 32 + 16 = 144 addresses, which fits within the 256-address /24 block.

| Department | Network / CIDR | Subnet mask | Usable range | Broadcast | Usable hosts |
|---|---|---|---|---|---|
| IT (50) | 192.168.10.0/26 | 255.255.255.192 | 192.168.10.1–62 | 192.168.10.63 | 62 |
| HR (30) | 192.168.10.64/27 | 255.255.255.224 | 192.168.10.65–94 | 192.168.10.95 | 30 |
| Finance (20) | 192.168.10.96/27 | 255.255.255.224 | 192.168.10.97–126 | 192.168.10.127 | 30 |
| Management (10) | 192.168.10.128/28 | 255.255.255.240 | 192.168.10.129–142 | 192.168.10.143 | 14 |

Unused address space: 192.168.10.144–192.168.10.255 remains available for growth, infrastructure or additional subnets. The usable host counts shown are capacity, not a recommendation to fill every address.

Design reasoning: /26 is the smallest standard subnet that accommodates 50 hosts; /27 accommodates 30 hosts exactly, and /28 accommodates 10 hosts with room for 4 additional usable addresses. Reserve gateway addresses consistently (for example, the first usable address) and document any DHCP exclusions.

# 6. Company Network Design Scenario

## Q12. Scenario: 45 employees, 5 servers and 10 network devices

Network type: a LAN within the office, with a WAN/internet connection through the ISP. The logical design separates employee, server and management traffic. The actual number of access switch ports should account for endpoints, access points, printers, servers, uplinks and future growth; a managed switch stack with adequate spare ports is preferable to choosing a switch solely from the number of employees.

## Proposed logical topology

| |
|---|
| **Internet / ISP** |
| ↓ |
| **Router + Firewall**<br>NAT • VPN if needed • traffic rules |
| ↓ |
| **Managed Core / Access Switches**<br>VLANs • port security • monitoring |
| ↓ |
| Employee PCs & Wi-Fi \| Server VLAN \| Network-management VLAN |

## Required components and roles

| Component | Role in the proposed network |
|---|---|
| ISP connection and router | Provides the upstream connection and routes traffic between the office LAN and external networks. |
| Next-generation or business firewall | Applies inbound/outbound rules, blocks unsolicited connections and logs security events. |
| Managed switches | Connect wired devices; support VLANs, access controls and monitoring. |
| Wireless access points | Provide staff Wi-Fi; use WPA2-AES/WPA3 and a separate guest network. |
| Five servers | Host required business services; place in a restricted server VLAN and expose only necessary services. |
| DHCP/DNS services | Provide IP configuration and name resolution; use trusted internal services or approved resolvers. |
| UPS and backups | Improve resilience and protect systems from short power interruptions and data loss. |

## Suggested segmentation

| Segment | Example purpose | Key rule |
|---|---|---|
| VLAN 10 — Employees | Staff computers and printers | Permit only approved business services; block direct access to management interfaces. |
| VLAN 20 — Servers | Five internal servers | Allow only required ports from approved clients; restrict outbound traffic where practical. |
| VLAN 30 — Network management | Switches, APs and administrator devices | Administrator-only access; use MFA where supported and log changes. |
| VLAN 40 — Guest Wi-Fi | Visitors' internet access | Internet only; deny access to internal subnets. |



**How an employee reaches a website**

The employee device obtains an IP address, subnet mask, default gateway and DNS settings through DHCP (or approved static configuration). DNS resolves the website name. Because the destination is external, the PC sends traffic to the default gateway. The router/firewall checks policy and routes permitted traffic to the ISP. The website responds through the established connection; the firewall tracks the session and the browser displays the response.

# 7. IP Troubleshooting Scenario

## Q13. Local communication works, but an external website does not

Given: the user has 192.168.1.25/24 and can communicate with another local computer. This suggests that the local interface and at least part of the LAN are functioning, but it does not prove that the default gateway, DNS or internet route is working.

### 1. Check link and interface

Confirm Wi-Fi/Ethernet is connected and the adapter is enabled. Check link lights or Wi-Fi association.

### 2. Verify IP configuration

On Windows, run ipconfig /all. Confirm 192.168.1.25, mask 255.255.255.0, a valid default gateway (for example, 192.168.1.1), and DNS server addresses. Check for duplicate IPs or an unexpected 169.254.x.x address.

### 3. Test the local TCP/IP stack

Ping 127.0.0.1, then ping 192.168.1.25. These tests help identify local TCP/IP or adapter issues.

### 4. Test another LAN host

Ping a known local computer or server. Since local communication already works, compare the result with the expected network configuration.

### 5. Test the default gateway

Ping the configured gateway. If it fails, inspect the client VLAN, cable/Wi-Fi, switch port, gateway address and firewall/ICMP policy. A failed ping alone is not conclusive because ICMP may be blocked.

### 6. Test internet reachability by IP

Try a known permitted external IP address, for example ping 1.1.1.1, where ICMP is allowed. If IP connectivity works but names fail, suspect DNS; if it fails, check router, ISP link and firewall rules.

### 7. Test DNS

Run nslookup example.com. If resolution fails, verify DNS server settings, DNS service availability and whether UDP/TCP port 53 is permitted. Compare with an approved alternate resolver if policy allows.

### 8. Check route and browser-specific causes

Run tracert example.com on Windows, inspect VPN/proxy settings, system date/time, browser error details and whether other websites work. Some networks block traceroute or ping.

### 9. Check router and ISP

Confirm the router has a valid WAN address, default route and working ISP connection. Review firewall logs, NAT, DNS forwarding and any outage notice.

### 10. Document and verify

Change one item at a time, record results, avoid disabling the firewall as a troubleshooting shortcut, and retest the original website.

### Useful Windows commands

```
ipconfig /all
ping 127.0.0.1
ping 192.168.1.25
ping 192.168.1.1
nslookup example.com
tracert example.com
```



Interpretation: if the gateway is reachable and an external IP is reachable but example.com does not resolve, DNS is the leading suspect. If the gateway cannot be reached, investigate local network configuration first. If the gateway works but external IP connectivity fails, investigate router forwarding, firewall policy, WAN connectivity and the ISP.

# 8. Security Recommendations

• **Identity and access:** use unique accounts, least privilege, strong passphrases and multifactor authentication for administrator and cloud accounts.

• **Secure configuration:** change default credentials, disable unused services and ports, use secure management protocols, and maintain configuration backups.

• **Patch management:** apply supported firmware and operating-system updates after appropriate testing.

• **Network controls:** segment employees, servers, guests and management devices; permit only necessary traffic through firewall rules.

• **Wireless security:** use WPA2-AES or WPA3, strong Wi-Fi credentials, and isolated guest access. Avoid obsolete WEP and open staff networks.

• **Data protection:** encrypt sensitive data, maintain tested offline or immutable backups, and limit access to business need.

• **Monitoring and response:** review firewall and server logs, synchronize time, document incidents, and define escalation contacts.

• **User awareness:** train staff to identify phishing, report suspicious messages and verify unusual payment or credential requests.

# 9. Conclusion

This project demonstrates how cybersecurity principles and networking fundamentals work together. The CIA Triad frames security goals; switches, routers, firewalls and protocols enable communication; CIDR calculations define efficient IP networks; and segmentation plus a structured troubleshooting process improve security and reliability. The proposed office design gives each department a suitable subnet, keeps management and guest traffic separated, and provides a clear path for investigating internet-access failures.

# 10. References

1. Unlox Academy, *Minor Project Assessment: Cybersecurity & Networking Fundamentals*, provided course document (assessment topics and marking scheme).

2. IETF, RFC 1918 — Address Allocation for Private Internets. https://www.rfc-editor.org/rfc/rfc1918

3. IETF, RFC 4632 — Classless Inter-domain Routing (CIDR): The Internet Address Assignment and Aggregation Plan. https://www.rfc-editor.org/rfc/rfc4632

4. IETF, RFC 8200 — Internet Protocol, Version 6 (IPv6) Specification. https://www.rfc-editor.org/rfc/rfc8200

5. NIST, Cybersecurity Framework (CSF) 2.0. https://www.nist.gov/cyberframework

## Submission checklist

• ■ All assessment questions answered

• ■ Subnet calculations and working included

• ■ Network flow and topology diagrams included

• ■ Scenario answers include reasoning

• ■ Student details completed on the cover page

• ■ PDF reviewed for formatting before upload